



General results of work on Metallogenic map of Northern, Central, and Eastern Asia

A.G. Pakhalko, E.M. Pinsky, G.A. Shatkov



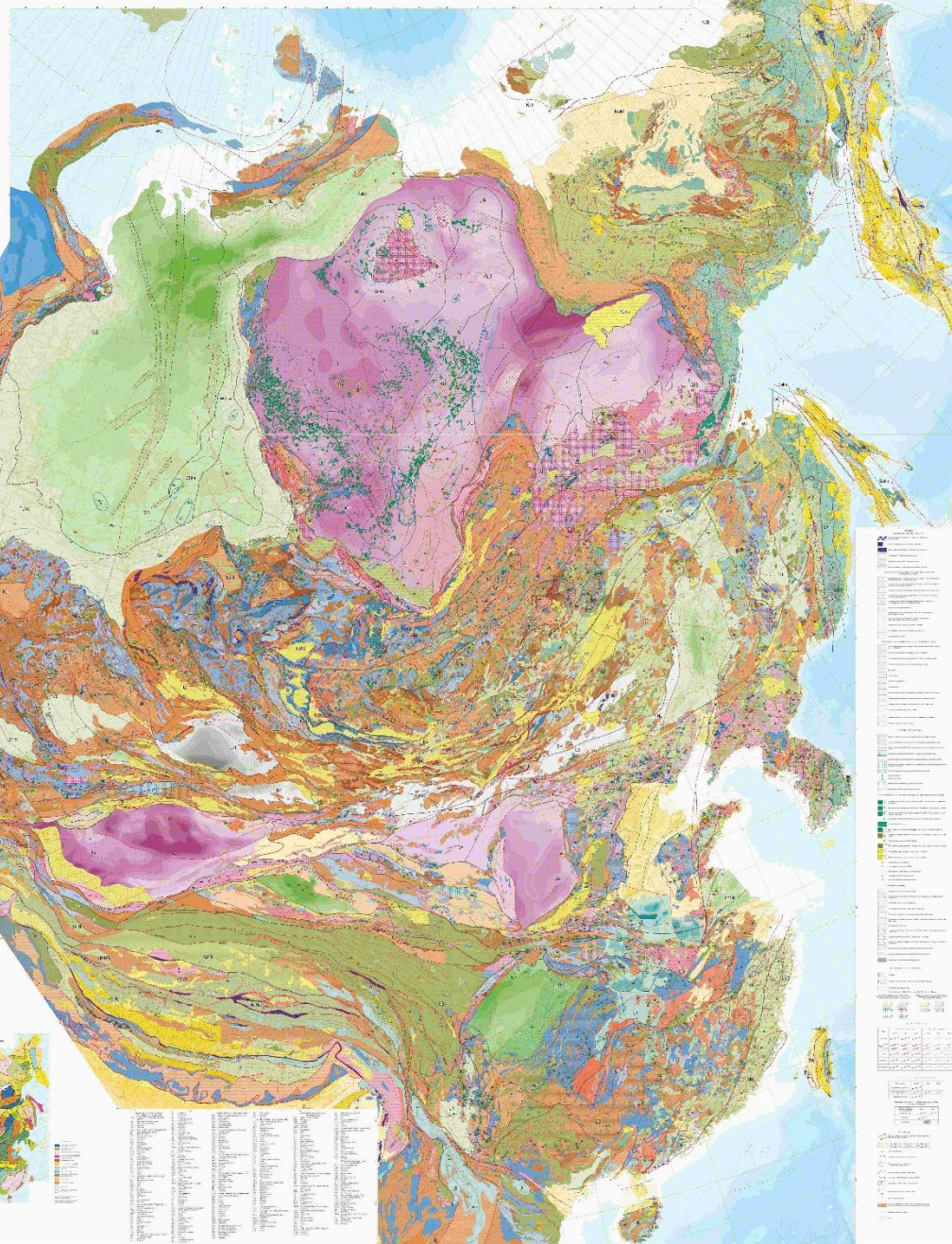
METALLOGENIC MAP OF NORTHERN-CENTRAL- EASTERN ASIA AND ADJACENT AREAS

ATLAS OF GEOLOGICAL MAPS OF NORTHERN-
CENTRAL-EASTERN ASIA AND ADJACENT AREAS

1:2 500 000

METALLOGENIC MAP OF NORTHERN CENTRAL-
EASTERN ASIA AND ADJACENT AREAS

1:2 500 000



Metallogenic map of the Central, Northern and Eastern Asia scale 1:2.5 M

Coordinating Board: Dong Shuwen, Li Tingdong (China), B. Uzhkenov, A. Kiselev (Kazakhstan), Kim Bok Chul (Republic of Korea), O. Chuluun, D. Javkhlanbold, (Mongolia), A. Morozov, O. Petrov (Russia)

Editors-in-Chief: O. Petrov, A. Morozov, E. Kiselev, S. Shokalsky (Russia), Dong Shuwen, Chen Tingyu (China), O. Chuluun, O. Tomurtogoo, (Mongolia), B.S. Uzhkenov, M. Sayduakasov (Kazakhstan), Hwang Jae Ha, Kim Bok Chul (Republic of Korea)

Editors: G. Shatkov, S.Shokalsky (Russia), Chen Tingyu, Geng Shufang (China), G. Dedjidmaa O. Tomurtogoo (Mongolia); Se-Jung Chi (Republic of Korea)

Authors: G. Shatkov, E. Pinsky, N. Solovyev, V. Feoktistov, V. Shatov, L. Rucheikova, V. Guschina, Yu Mironov, A. Molchanov, V. Masaytis, V.Puring, L. Krasny, A.Volsky, B. Blyuman, A. Sobolev, N. Gusev, V. Shamakhov, V. Stromov, V. Proskurnin, V. Shpikerman, A. Firsov, S. Strelnikov (Russia); Chen Tingyu, Geng Shufang, Dong Shuwen, Chen Binwei, Hao Meiying, Huang Dianhao, Song Tianrui, Sheng Jifu, Zhu Guanxiang, Yan Keming, Min Longrui, Jin Ruogu, Liu Ping, Fan Benxian, Ju Yuanjing, Wang Zhenyang, Han Kunying, Wang Liya (China); G. Dedjidmaa, O. Tomurtogoo, B. Delgertsogt, Ts. Enkhbat (Mongolia), Kim Bok Chul, Hwang Jae Ha (Republic of Korea); A.L. Kiselev, O.Fedorenko, L.A. Miroshnichenko, D.V. Gurevich (Kazakhstan)

Data contributors: A. Afanasiev, A. Belyaev, S. Buzovkin, S. Kashin, S. Kalabashkin, Yu. Bogdanov, V. Chernov, A. Korshunov, G. Korshunova, N. Vlasov, V. Prozorovsky, E. Plyushchev, A. Melgunov, P. Paderin, B. Markovsky, A. Ioganson, A. Stukanov, G. Shor, Yu. Shuvalov (Russia); Chen Zuyi, Zhao Fengmin (China); A. Rustamov, I. Golovanov, R. Mirkamalov, B. Sakhadzhaev, Yu Ezhkov (Uzbekistan); R. Dzhenchuraeva, A. Bakirov, Sh. Murzagaziev (Kyrgyzstan); V. Minaev (Tadjikistan)

Digital cartography: L. Rucheikova, A. Gureev, M.Demidovich, E. Leontieva, M. Semenova, E. Reneva, M. Shipova, M. Kulchitskaya, V. Guschina, I. Bykova, S. Rebrov, N. Seregina (Россия); Fan Benxian, Ju Yuanjing, Wang Liya, Xie Liangzhen (China) B.Zul, Ts. Minjinsor (Mongolia)

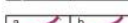
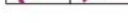
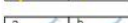
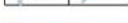
Legend of Metallogenic map of the Central, Northern and Eastern Asia scale 1:2.5 M

Mineragenic zoning

Boundaries of metallogenic units of different rank metallogenic

-  Mineragenic provinces
-  Mineragenic areas and zones

Boundaries of ore zones, regions (a); knots (b)^{1,2}

- | | |
|--|---|
|  | Ferrous metals |
|  | Colored or Nonferrous metals |
|  | Rare metals |
|  | Rare metals, trace elements and Rare Earth Elements |
|  | Radioactive metals |
|  | Precious metals |
|  | Diamonds, precious and ornamental stones |
|  | Agro raw material |
|  | Mining chemical and mining technical raw materials |
|  | Optical materials |
|  | Salts |

List of indexes:

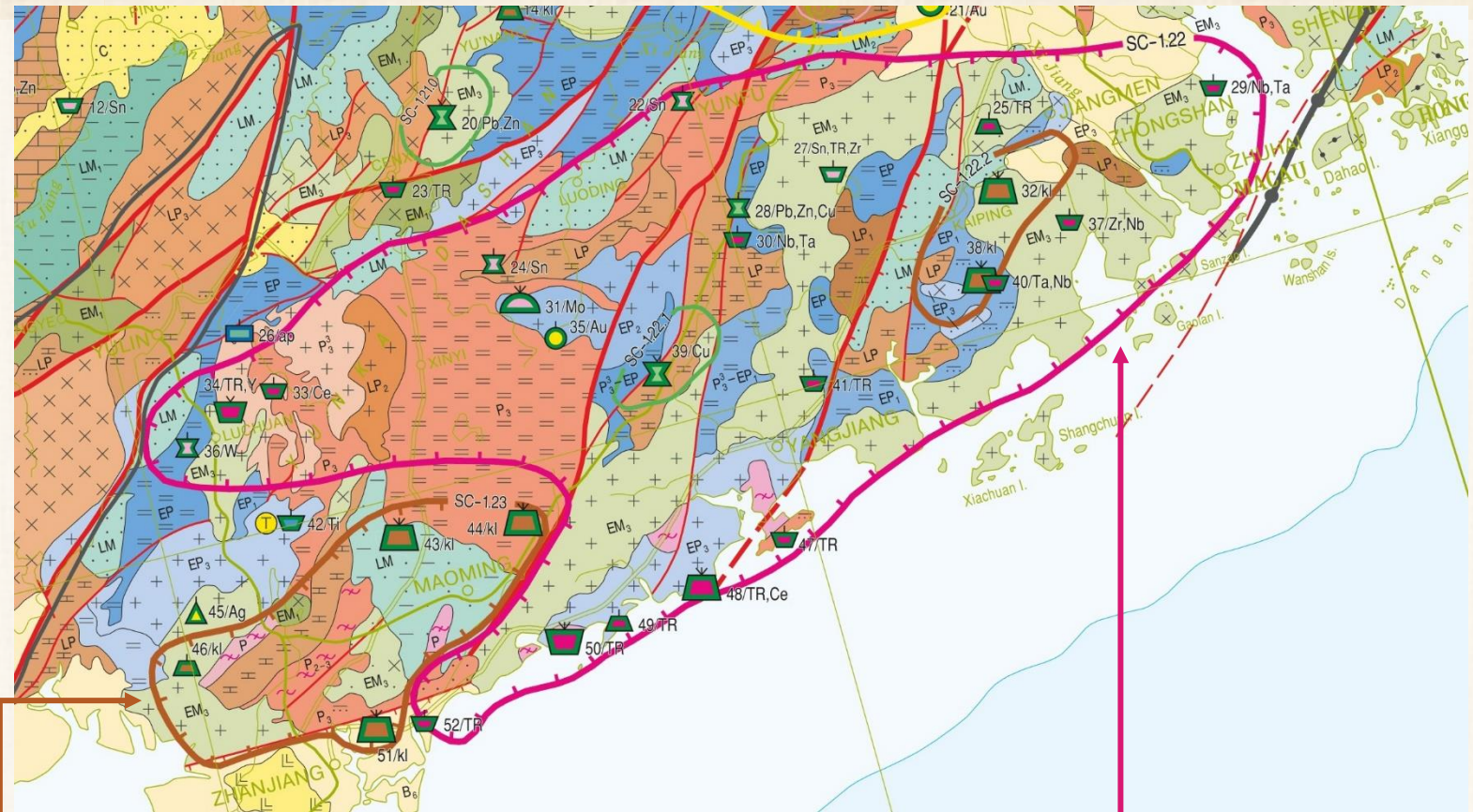
1 - The color reflects the metallogenic specialization, the group of the mineral deposits

2 - Thickness of the lines shows the production of the ore zones, regions and knots:

 - high,  - medium,  - low, unspecified,  - predicted.

Metallogenic unit symbols on the Map

- KA - Metallogenic province
- KA-1 - Metallogenic areas and zones
- KA-1.1 - ore zones and regions
- KA-1.1.1 - ore knot
- KA-1.1.0 - ore knot out of ore zones and regions



SC – South China metallogenic province

SC-1 – Luoxiaoshan-Wuyishan-Nanling metallogenic zone

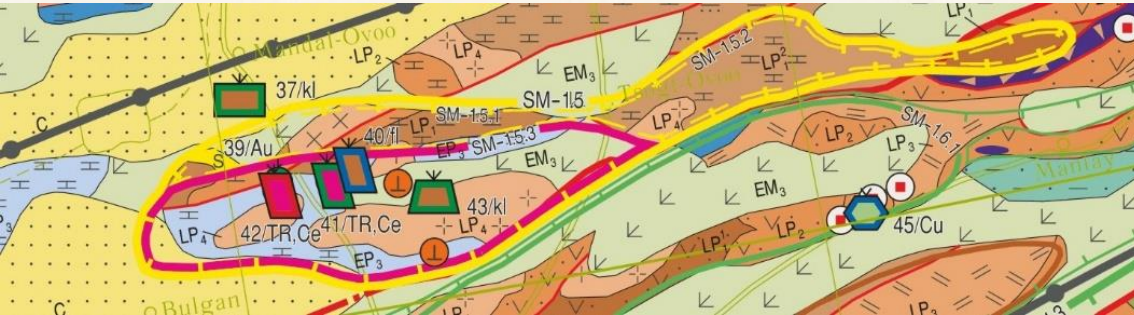
SC-1.22 – Luchuan-Xinyi-Zhongshan ore zone

SC-1.22.1 – Yangchun ore knot

SC-1.22.2 – Kaiping ore knot

SC-1.23 – Maoming-Zhanjiang ore zone

Legend of Metallogenic map of the Central, Northern and Eastern Asia scale 1:2.5 M



SM-1.5.3 – Mushgai-Khudag ore knot



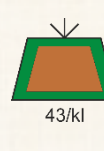
Mushgai-Khudag



Bayanjhushuu

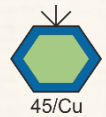


Avdarant



Tsogt-Ovoo

SM-1.6.1 – Kharmagtai ore knot



Kharmagtai



Mandal-Ovoo

37/kj

Mineral deposits

Groups of Minerals

- Ferrous metals (Fe, Mn, Cr, Ti, V)
- Colored or Nonferrous metals (Cu, Pb, Zn, Ni, Co, Al)
- Rare metals (Sn, Mo, W, As, Sb, Hg, Bi)
- Rare metals, trace elements and Rare Earth Elements (Ta, Nb, Zr, Y, TR, Li, Cs, Be, Sc, Ga, Ge, Re, Gd, In)
- Radioactive (U)
- Precious metals (Au, Ag, Pt, PG)
- Diamonds, precious and ornamental stones¹
- Agro raw material²
- Mining chemical and mining technical raw materials³
- Optical materials⁴
- Salts (Na.s, K.s, Mg.s)

List of indexes:

1 - diamond - di, amethyst - amt, agalmatolite - agal, beryl - be, turquoise - tr, demantoid - dem, jadeite - jad, emerald - em, lazurite - lz, nephrite - np, pirope - pir, rhodonite - rho, sillimanite - sil, spinel - sp, topaz - to, tourmaline - tu, chalcodony (agate, carnelian) - ch, chrysolite - chy, chrysopras - chr, chrome diopside - cdi, charoite - cha, jasper - jas,

2 - apatite - ap, phosphorite - ph;

3 - asbestos - asb, barite - ba, bentonite - bn, brucite - brs, vermiculite - ver, gypsum - gyp, graphite - grf, kaolin - kl, magnesite - mg, muscovite - mus, ophalcite - oph, synnyrite - syn, strontianite - str, zeolite - zeo, talc - tl,

4 - iceland spar - is, piezooptical quartz - qz

Ore quality of deposits (outline colour of the symbol)

	Rich
	Normal
	Poor
	Undefined

Genetic types and sizes of deposit marks

Genetic type			Large	Medium	Small
M a g m a t i c	In layered ultramafic-mafic and alpine-type ultramafic complexes	LUMC			
	Ophiolitic suites	OPH			
	Foidites and foidolites, carbonatites	FAC			
	Kimberlites and lamproites	KIM			
Plutogenic hydrothermal and metasomatic rocks	Pegmatite	PEG			
	Greisen and feldspatic in granite and leucogranite	GAF			
	Skarn	SK			
	Telethermal and epithermal	TLT			
	Apogranite	AG			
	Vein and stockwork	VS			
	Porphyry	PORF			
Volcanogenic and volcanogenic-sedimentary hydrothermal	In volcano-tectonic structure and basement rocks	HPV			
	Volcanogenic and volcanogenic-sedimentary massive sulfide	VMS			
	Stratiform in volcanogenic-sedimentary rocks	SVS			
	Stratiform in carbonate rocks	MVT			
	Sedimentary-exhalation in terrigenous and terrigenous-carbonate rocks	SEDEX			
Exogenetic-epigenetic	Hydrogenous infiltrational and exfiltrational	HIE			
	Sedimentary	SED			
	Placer	PL			
	Waste mantle	WM			
Regional or locally transformed	Metamorphic	MET			
	Polygenetic	POL			
	Impact	IMP			



ATLAS OF GEOLOGICAL MAPS
OF NORTHERN-CENTRAL-EASTERN
ASIA AND ADJACENT AREAS

METALLOGENY OF NORTHERN, CENTRAL AND EASTERN ASIA

Saint-Petersburg 2018

Explanatory note for Metallogenic map of Northern, Central and Eastern Asia

Editors-in-Chief: O.V. Petrov, A.F. Morozov, E.A. Kiselev, S.P. Shokalsky (Russia), Dong Shuwen, Chen Tingyu (China), O. Chuluun, O. Tomurtogoo (Mongolia), B.S. Uzhkenov, M.A. Sayduakasov (Kazakhstan), Hwang Jae Ha, Kim Bok Chul (Korea)

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Revision of Explanatory note and metalogenic units for Metallogenic map of Northern, Central and Eastern Asia



Chinese part

Supervisor: Chen Tingyu

Revised: **content (partly and fully rewritten text, updated ages for deposits), grammar, structure, references**

The sedimentary Fe deposits in the ore region occur mainly in Jurassic rocks, mostly small in size.

Gongxian-Junlian iron ore region (UY-1.4; 5,964.95 km²)

The sedimentary Fe deposits in the ore region occur in Triassic rocks, mostly small in size.

Wanzhou salt ore knot (UY-1.5.0; 745.63 km²) - The *Gao'feng* salt deposit in Wanzhou is a medium-sized deposit formed in the Early-Middle Triassic.

Kangdian metallogenic area (UY-2)

123,246 km²

In tectonics, the Kangdian metallogenic area is mainly situated in the Panxi (west Panzhihua)-Dianzhong(central Yunnan) Variscan rift zone. More than 300 intrusive rock bodies are distributed in this area, formed in the period from the Early Variscan to the Late Yanshanian. These rock bodies are mafic-ultramafic or granitoid in composition. Minerals related to mafic-ultramafic rocks are Fe, Ti, V, Cu, Ni, Co, Pt, Pd, serpentine and asbestos. The intrusive rock bodies related to mineralizations are mostly emplaced in country rocks of the Proterozoic and Paleozoic. The deposits are mainly magmatic or, subordinately, hydrothermal in origin.

Panzhihua-Xichang iron-titanium-vanadium ore zone (UY-2.1; 11,346.83 km²)

The V-bearing titanomagnetite deposits related to Variscan mafic and ultramafic rocks are widely distributed in this ore zone. Ore-bearing rock zones are constrained by several deep fractures in the N-S direction, *i.e.*, the Anninghe fractures, the Mopanshan-Xigeda fractures, and the Panzhihua fractures. This ore zone stretches in the N-S direction for 300 km in length and 10 ~15 km in width. More than 10 medium and large ore deposits are explored in this zone. Thereinto, *Panzhihua*, *Hongge*, *Zhonggan'gou* and *Zhongliangzi* in the Panzhihua area, *Taihe* in the Xichang area, and *Baima* in the Miyi area are representatives. The ores contain not only Fe, Ti and V, but also Cr, Ni, Cu, Co, Mn, Ga, Sn, Te, Sc and Pt group elements.

Volcano-sedimentary Fe deposits of the Mesoproterozoic are distributed in the south section of the ore zone; they are generally small in size, which include *Kuangchanliangzi* and *Niuchang* in Yanyuan.

Yuan Haihua reported a Rb-Sr isochron age of 343.48 Ma for The Hongge Rock Body in 1985 [See: Zhang Yunxiang (ed.), 1985], and Institute of Geology, Chinese Academy of Sciences, published whole rock isochron ages of 380Ma and 310.5 Ma for the Panzhihua and Hongge Rock Bodies in 1988 [See: Cong Bolin (ed.), 1988], indicating the Devonian-Carboniferous age of the mafic-ultramafic rock bodies and relevant Fe mineralization.

Huili copper-nickel ore region (UY-2.2; 2,055.87 km²)

Three ore deposits are present in this ore region, they are the *Lalachang* Cu deposit, the *Datongchang* Cu deposit and the *Limahe* Ni deposit. The *Lalachang* Cu deposit is volcano-sedimentary in origin, medium in size; besides Cu, the ores contain Mo. The *Datongchang* Cu deposit is a sedimentary deposit. The *Limahe* Ni deposit is a small deposit occurring in mafic and ultramafic rocks. Tang Zhongli *et al.* [1993] reported K-Ar ages of 353~332 Ma for the pyroxene from *Limahe*.

Yuanmou platinum ore region (UY-2.3; 3,286.44 km²)

The *Zhubu* Pt-Pd ore deposit in Yuanmou occurs in peridotite-pyroxenite-gabbro. Beside Pt and Pd, this deposit contains Ni, Au, Ag, Os, Ir and Ru. It is a small deposit.

The *Reshuitang* Pt deposit in Yuanmou occurs in peridotite. The ores contain Pt, Pd, Cu and Ni. It is also small. The *Anyi* Pt deposit is similar, the ores contain Pt, Pd, Ti and V.

Moreover, a graphite deposit is present in Mouding of this region, *i.e.*, the *Shujie* graphite deposit. This is a metamorphic deposit, medium in size.

Xinping copper-iron ore region (UY-2.4; 4,892.46 km²)

In the Amu Darya, syncline there is a thick halogen strata, which is likely to contain specific sodium and other salt deposits. These units are not shown on the Metallogenic Map. There is also information on phosphate occurrences in the south-western part of the area.

Southern Tien Shan metallogenic province (ST), Uzbekistan, Kyrgyzstan, Tajikistan, China
280,000 km²

The Southern Tien Shan (Tianshan) metallogenic province is the eastern extension of the Kyzylkum metallogenic province. In tectonics, it is sandwiched between the Middle Tien Shan (Tianshan) in the north, and the Tarim block in the south. This metallogenic province consists of some parts in Central Asia and some parts in China with a total length of 2,000 km from west to east in the strike direction.

Different from the Northern Tien Shan (Tianshan) and the Middle Tien Shan (Tianshan), the Southern Tien Shan (Tianshan) consists of Middle Paleozoic rocks formed in geological settings from mid-ocean ridge to passive continental margin. Variscan overthrusting and folding along with the tectono-magmatism were quite severe. There exists a difference in the crustal composition between the northern part and the southern part of this metallogenic province. In the northern part, the basement is composed of granite-amphibolite, while in the southern part, the basement is composed of granite-gneiss [Baimukhamedov *et al.*, 1985, *reference?*]. The northern part is dominated by clastic sediments; basaltic volcanic rocks and granitoids are widely spread there. In the southern part, carbonate rocks and more acidic volcanic rocks and granitoids are widely distributed

Main minerals of the province include Hg, Al, fl, Au, Pb, Sb, Sn, W, U. As opposed to Kyzylkum province the area in question is characterized by abundance of small deposits. There are 233 deposits in the province, 8 are large: *Khaidarkan* (Hg, Sb, fl), ST-2.3.0, Kara-Archa (Hg, Sb), ST-2.3.0, *Dzhizhikrut* (Sb, Hg), ST-1.3.1, *Chonkoy* (Hg), ST-2.5.0, *Kadanjai* (Sb, Hg), ST-2.4.0, *Dshanguz-Bulak* (Hg, Sb, fl, As) ST-2.3.0, *Botzzygoer* (Zr, Nb, TR), ST-4.2, *Fanchankou* (V), ST-5.2.1.

Many ore knots are located within the Late Hercynian ore belt of the northeast strike.

Primary mineralization age of the province, except sedimentary deposits of lead and zinc (D₂), bauxite (C₂), copper (K₁₋₂) and strontium (K₂), deposits of Au, Sn, and Sb, Hg, is closely

Revision of **Explanatory note** and **metallogenic units** for Metallogenic map of Northern, Central and Eastern Asia



Supervisor: Pakhalko Aleksey, Shatkov Georgy
Revised: grammar, structure

uranium deposit. In the Southern Novaya Zemlya area there are also stratiform Paigotinskoe-4 and Razdelnoe Pb-Zn deposits. These deposits **generate-form** the Sakhaminsky ore region. Veinlet-disseminated calcite-fluorite ores of the *Chernogubsky* ore knot, which resources are estimated at 20 million tons of CaF_2 are located in the D₁ limestone.

Fluorite deposits (*Amdermu*) are related to the Late Triassic-Early Jurassic (collision) stage of evolution. The issue concerning the age and genesis of the deposit is not understood. Perhaps it formed during **riftogenesis-that-occure-drifting** at the early stage of formation of the West Siberian plate (T₁).

In general, lead and zinc ores, fluorite, manganese as well as gold, copper, uranium, **seek crystal**-agate deposits are currently of prime importance in the Paikhoi-Novaya Zemlya province; phosphate and bauxite ores are promising. The Kara and Ust-Kara astroblemes remain poorly understood.

Taimyr-Severnaya Zemlya province (TNL), Russia 565,000 km²

The province is confined to the **self-titled** folded region-**of-the-same-name**, which forms part of the Hercynian-Early Cimmerian basement of the Barents-Northern Kara young platform [Proskurnin 2013]. Main minerals are Au, Pb, Zn, Cu, Ni, Mo, U, grf. The province hosts twelve ore deposits and many occurrences, but large ore deposits have not been found yet.

As to the time of consolidation and metallogenic orientation, the province is subdivided into three metallogenic areas: Northern Taimyr-Severnaya Zemlya (TNL-1), Mid-Taimyr (TNL-2) and Southern Taimyr (TNL-3).

Northern Taimyr-Severnaya Zemlya area (TNL-1)

In-the-area-4 there are five **small** ore deposits, **but-large ore**-deposits have not been found. Main minerals are Au placers (Golyasheva River, Tora River, Studenaya River). There **are-no-large ore-knots; they only** NQ Golyashevsky **medium size** ore **knot is-of-medium-size**. Source-rocks-for **of gold are-is** hydrothermally **occurences-in-altered** metamorphosed sedimentary rocks. **High** **P**potential gold ore knots-**have-been-outlined**: Chukchi-Oleninsky (TNL-1.4.0), Konezhninsky (TNL-1.6.0), etc.

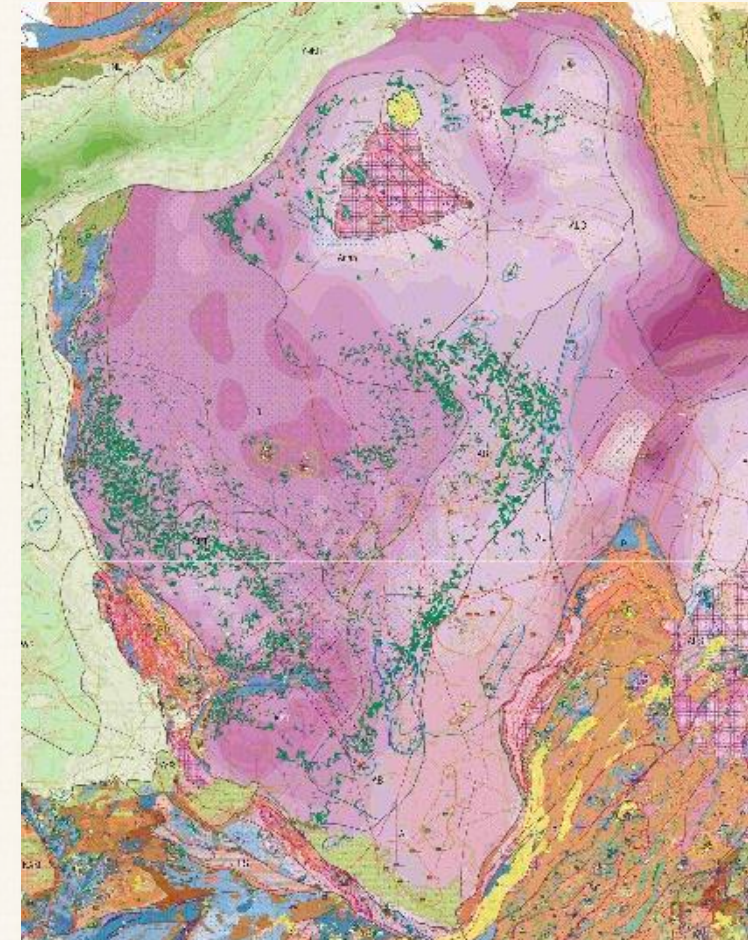
In the Northern Taimyr-Severnaya Zemlya subprovince, there are two metallogenic zones: Severnaya Zemlya zone (gold-iron-base metals) and Mininsky-Bolshevik zone (gold-rare metals) separated by the Main Severnaya Zemlya Fault.

Russian part

Supervisor: Gusev Nikolay

Revised: deposits

Western part of **Siberian platform**



Added or updated data for 31 deposits

The flag of Russia, featuring three horizontal stripes of white, blue, and red, is displayed within a circular frame.

Supervisor: Shpikerman Vladimir
Revised: deposits and metallogenic units
Northeastern Russia

Revised: deposits and metallogenic units

Northeastern Russia

Original

Oloi metallogenic province
3 ore knots and 4 deposits

Borders of Kolyma, Verkhoyansk and
Okhotsk metallogenic provinces
6 ore zones, 15 ore knots and 9 deposits

Revision of Explanatory note and
metallogenic units
for Metallogenic map of Northern,
Central and Eastern Asia



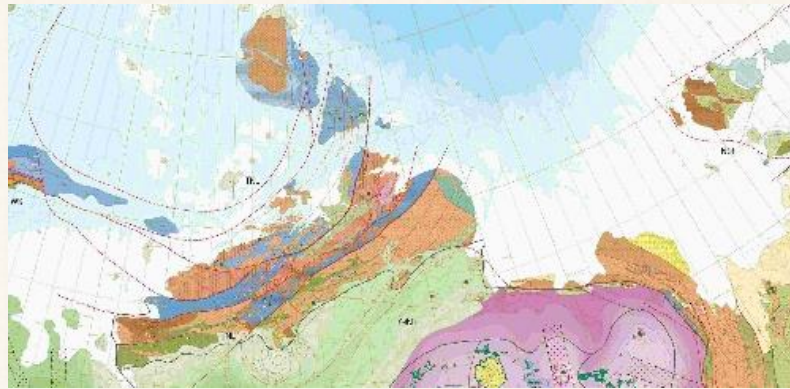
Russian part

Under revision

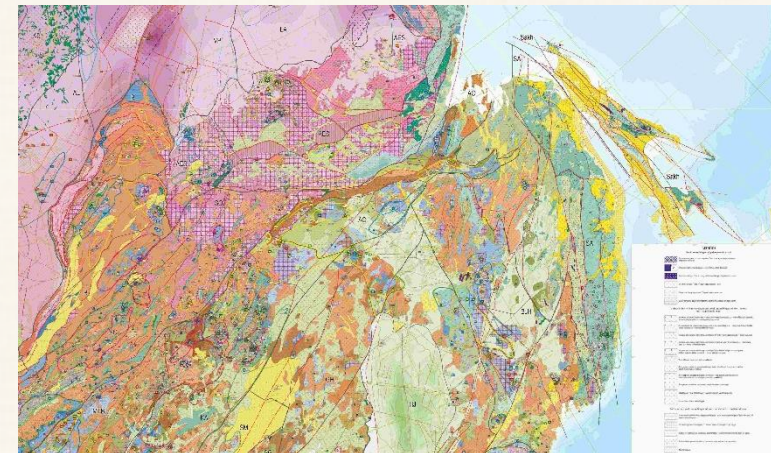
Arctic regions, Ural mountains
Southeastern Russia, Kamchatka and Chukotka
peninsula



Ural mountains



Arctic regions



Southeastern Russia



Kamchatka and
Chukotka
peninsula

Metallogenic Map of Northern, Central, and Eastern Asia

Analysis and compilation of data from [database](#)

<i>Item No.</i>	<i>Field Description</i>	<i>Field Name</i>
1	Unique ID of the object	ui
2	Deposit name	dn
3	Synonym for the deposit name	sdn
4	Index of deposit on the map	din
5	Country	con
6	Administrative binding	adm
7	Coordinates XX° N, latitude (decimal degrees)	lt
8	Coordinates XX° E, longitude (decimal degrees)	lg
9	Binding to sheets, scale 1:1,000,000	st
10	Group of minerals	gm
11	Main minerals	mm
12	Associated minerals	am
13	Genetic type	gt
14	Geological and commercial type	ct
15	Deposit spread	ds
16	Ore quality	oq
17	Morphology of ore bodies	mor
18	Age of mineralization	age
19	Methods of mining	met
20	Degree of development	sta
21	Ore region, area	are
22	Geological formation conditions	geod

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Ferrous Metals (fm)
<i>Iron (Fe)</i>
<i>Manganese (Mn)</i>
<i>Chrome (Cr)</i>
<i>Titanium (Ti)</i>
<i>Vanadium (V)</i>
Non-Ferrous Metals (nm)
<i>Copper (Cu)</i>
<i>Nickel (Ni), Cobalt (Co)</i>
<i>Lead (Pb), Zinc (Zn)</i>
<i>Aluminium (Al)</i>
Rare Metals (rm)
<i>Tin (Sn)</i>
<i>Molybdenum (Mo)</i>
<i>Tungsten (W)</i>
<i>Antimony (Sb)</i>
<i>Mercury (Hg)</i>
<i>Bismuth (Bi)</i>
<i>Arsenic (As)</i>
Rare, Scattered and Rare Earth Elements (rmm)
<i>Tantalum (Ta), Niobium (Nb)</i>
<i>Lithium (Li)</i>
<i>Cesium (Cs)</i>
<i>Beryllium (Be)</i>
<i>Germanium (Ge)</i>
<i>Rare Earth Elements (TR, Zr, Hf)</i>
Precious Metals (prm)
<i>Gold (Au)</i>
<i>Silver (Ag)</i>
Platinum Group Metals (PG)
Radioactive Elements (rad)
<i>Uranium (U)</i>

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20	Degree of development	sta
21	Ore region, area	are
22	Geological formation conditions	geod

Mining Chemical and Engineering Materials (mtm)
<i>Fluorspar (fl)</i>
<i>Boron (B)</i>
<i>Sulphur (S)</i>
<i>Barite (ba)</i>
<i>Graphite (grf)</i>
<i>Brucite (brs)</i>
<i>Asbestos (asb)</i>
<i>Plaster (gyp)</i>
<i>Muscovite (mus)</i>
<i>Phlogopite (phl)</i>
<i>Vermiculite (ver)</i>
<i>Kaolin (kl)</i>
<i>Bentonite (bn)</i>
<i>Zeolite (zeo)</i>
<i>Magnesite (mg)</i>
<i>Quartzite (qua)</i>
<i>Perlite (per)</i>
<i>Synnyrite (syn)</i>
<i>Talc (tl)</i>
Agricultural Materials (agr)
<i>Apatite (ap)</i>
<i>Phosphorite (ph)</i>
Salts (sal)
<i>Sodium salt (s.Na)</i>
<i>Potassium salt (s.K)</i>
Diamonds, Precious, Semiprecious and Ornamental Stones (dia)
<i>Diamonds (di)</i>
<i>Precious, Semiprecious and Ornamental Stones (dia)</i>
Optical Materials (om)
<i>Optical quartz (gz)</i>
<i>Iceland spar (is)</i>

Metallogenic Map of Northern, Central, and Eastern Asia

Analysis and compilation of data from [database](#)

Item No.	Field Description	Field Name
1	Unique ID of the object	ui
2	Deposit name	dn
3	Synonym for the deposit name	sdn
4	Index of deposit on the map	din
5	Country	con
6	Administrative binding	adm
7	Coordinates XX° N, latitude (decimal degrees)	lt
8	Coordinates XX° E, longitude (decimal degrees)	lg
9	Binding to sheets, scale 1:1,000,000	st
10	Group of minerals	gm
11	Main minerals	mm
12	Associated minerals	am
13	Genetic type	gt
14	Geological and commercial type	ct
15	Deposit spread	ds
16	Ore quality	oq
17	Morphology of ore bodies	mor
18	Age of mineralization	age
19	Methods of mining	met
20	Degree of development	sta
21	Ore region, area	are
22	Geological formation conditions	geod

Genetic types and sizes of deposit marks					
Genetic type			Large	Medium	Small
M a g m a t i c	In layered ultramafic-mafic and alpine-type ultramafic complexes	LUMC			
	Ophiolitic suites	OPH			
	Foidites and foidolites, carbonatites	FAC			
	Kimberlites and lamproites	KIM			
Plutogenic hydrothermal and metasomatic rocks	Pegmatite	PEG			
	Greisen and feldspatic in granite and leucogranite	GAF			
	Skarn	SK			
	Telethermal and epithermal	TLT			
	Apogranite	AG			
	Vein and stockwork	VS			
	Porphyry	PORF			
Volcanogenic and volcanogenic-sedimentary hydrothermal	In volcano-tectonic structure and basement rocks	HPV			
	Volcanogenic and volcanogenic-sedimentary massive sulfide	VMS			
	Stratiform in volcanogenic-sedimentary rocks	SVS			
	Stratiform in carbonate rocks	MVT			
	Sedimentary-exhalation in terrigenous and terrigenous-carbonate rocks	SEDEX			
Exogenetic-epigenetic	Hydrogenous infiltrational and exfiltrational	HIE			
	Sedimentary	SED			
	Placer	PL			
	Waste mantle	WM			
Regional or locally transformed	Metamorphic	MET			
	Polygenetic	POL			
	Impact	IMP			

Metallogenic Map of Northern, Central, and Eastern Asia

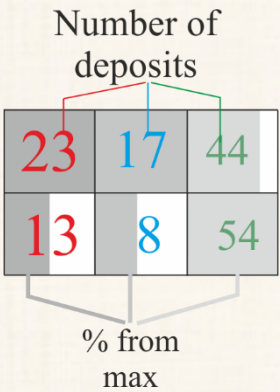
Analysis and compilation of data from [database](#)

<i>Item No.</i>	<i>Field Description</i>	<i>Field Name</i>
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18	Age of mineralization	age
19	Methods of mining	met
20	Degree of development	sta
21	Ore region, area	are
22	Geological formation conditions	geod

1	N-Q
2	K ₂ -E
3	J ₃ -K ₁
4	J ₂₋₃
5	PZ ₃ -T ₁
6	PZ ₂
7	PZ ₁
8	NP-Є ₁
9	PP ₄ -MP
10	PP ₁ -PP ₃
11	MA-NA

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis and data compilation from [database](#)



Ferrous																
Mineral		Fe			Mn			Cr			Ti			V		
Size of deposits		L	M	S	L	M	S	L	M	S	L	M	S	L	M	S
Total number of deposits		90	172	430	7	20	51	1	5	5	15	21	21	9	5	9
1	N-Q		1	5			8		1	1	1	6	4			
2	K ₂ -E	6	4	21		3	2			1	5	10	9			
3	J ₃ -K ₁	5	27	26			1									
4	J ₂₋₃	1	10	26			5				1		2			
5	PZ ₃ -T ₁	11	19	69	1	3	9					1	1			1
6	PZ ₂	18	37	119	1	3	7	1	3	1	1	1	2	5	2	
7	PZ ₁	11	7	48	1	5	9							4	2	8
8	NP-Є ₁	3	17	37	4	6	7		1	2	2	1				
9	PP ₄ -MP	8	4	26			1				2		3		1	
10	PP ₁ -PP ₃	8	9	21			2				1	1				
11	MA-NA	19	38	32							2	1				

Nonferrous																	
Cu			Pb			Zn			Ni			Co			Al		
L	M	S	L	M	S	L	M	S	L	M	S	L	M	S	L	M	S
48	96	230	29	91	323	13	18	47	5	8	29	8	8	21	29	51	116
		5									2			1			
1	2	8		2	2						8	1	1	4		7	22
5	3	18	5	12	58	2	1	2		1	4		1	2	1	12	18
3	5	6		8	28	1	5	29		4	11	2	4	11		3	1
12	18	64	1	13	80			2	3			3	1	2	10	18	48
22	48	91	15	45	129	8	7	10		2	1	1	1	1	17	9	26
1	14	30	2	6	25	2	3	2								1	
	1	6	2	1	1		1	1	1						1		
	1	3	3	2			1	1			2						1
4	3	2	1	2						1	1					1	
	1								1								

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis of spatial data and database

Number and specialization of **large** deposits within structures of different age of consolidated crust

1	By the beginning of MP
2	By the end of NP
3	By the middle of Devonian (Caledonides)
4	By Late Carboniferous - Permian (Variscans)
5	By the middle-late Triassic (Indosinides)
6	By the beginning of Cretaceous (Yanshanides)
7	By the end of the Early Cretaceous (Verkhoyanides)
8	Cenozoic (Himalaides)

Epochs	Total	LUMC	mm	OPH	mm	FAC	mm	KIM	mm	PEG	mm	GAF	mm	SK	mm	TLT	mm	AG	mm	VS	mm	PORF	mm	HPV	mm	VMS	mm
8	3																							1	S, Cu, Au		
7	16			1	Ti							2	Sn, W	3	W, B, Pb, Zn	2	Hg, W			5	Sn!, Au					1	Pb, Zn
6	45	2	Fe, Ti, ap			2	Y, Ce, ap, Fe					2	Sn, W	1	Sn, Cu, W					18	Sn!, Au, W	2	Cu, Sn, Mo, W	5	Au, Ag		
5	28	1	Cu, Co	1	asb	1	TR, Ce,			4	qz, Li, Sn, Be	1	W	1	Fe, Mo			1	Ta, Nb	1	Au	1	Mo	1	fl		
4	261	5	Ni, Co, Ti	10	asb, V, Ti, Cr,	4	TR, Ce, fl,			2	Ta, Nb	18	W, Mo, Be, fl	11	Fe, W, Cu, Au	9	Au, Hg, Sb	5	Ta, Nb, Zr	21	Au, Be, qz, Sn	11	Cu, Mo,	25	fl, U, Mo,	23	Cu, Pb, Zn
3	243	9	Fe, Ti, Ni, Co	8	asb, jad	9	Al, ap, TR			12	mus, Be, Li	12	W, Mo, Be	22	Fe, W, Au, Cu	5	Au, Ag, Al, fl	4	Zr, Y, Li, Ta,	23	Au!, U, Co, Ni	19	Mo, Cu,	6	zeo, Au,	6	Pb, Zn, Au
2	122	4	Fe, Ti, V	1	asb	6	Al, TR			1	Li			1	Cu	9	Hg!, Sb	1	Ta, Nb	14	Sn, Au, qz, Mo	3	Cu, Fe, Mo	2	Au	1	Cu, Fe
1	443	18	Cr, Ni, Co, Ti, V	2	Cr, tlk	17	Al, ap, Ta, Nb	12	Diam	18	mus, Ta, Nb	4	Sn, TR, W	21	Fe, Mo, Au	2	Qz	1	Ta, Nb	44	Au!, U!, Pb,	16	Mo, Cu	17	is, Au	2	Cu, Pb, Zn

Metallogenic evolution of Northern, Central, and Eastern Asia

Analysis using **Tectono-Metallogenic settings**

Deposits of different tectonic settings

Intraplate Tectonics and Magmatism					
LUMC	FAC	KIM	TLT	MVT	SEDEX

Accretion and Collision Structures						
OPH	GAF	SK	PORF	HPV	VMS	MET

	By the beginning of MP
	By the end of NP
	By the middle of Devonian (Caledonides)
	By Late Carboniferous - Permian (Variscans)
	By the middle-late Triassic (Indosinides)
	By the beginning of Cretaceous (Yanshanides)
	By the end of the Early Cretaceous (Verkhoyanides)
	Cenozoic (Himalaides)

		Accretion and Collision Structures	Intraplate Tectonics and Magmatism
1	N-Q	Au, Cu 2	
2	K ₂ -E	Cu, Mo, Au, Fe 7	1 Pb, Zn
3	J ₃ -K ₁	Mo, W, Au, Ag, Sn, U 69	6 Au, Fe
4	J ₂₋₃	Mo, W, fl, Cu, Au 34	11 TR, Au, Ag, fl
5	PZ ₃ -T ₁	W, Mo, Fe, grph, Cu 58	21 Co, Ni, Cu, Hg, Pb, Zn, Ti
6	PZ ₂	Cu, Fe, Pb, Zn, Au, Mo 74	48 Diam, Al, Pb, Zn, Fe
7	PZ ₁	Au, Zn, Mo, asb 17	10 Cr, Ti, Fe, diam, Zr, Ta
8	NP-Є ₁	Fe, grph, asb 12	9 Ti, Pb, Ta, Nb
9	PP ₄ -MP	Fe, apat 14	12 Apat, Pb, Zn, Ti, Ni, Cu
10	PP ₁ -PP ₃	grph, Fe, Cu, Pb, Zn 21	9 Ti, Fe, apat
11	MA-NA	Fe, grph, phl 28	3 Ni, Cu, Ti, apat

Iron deposits (Fe)

[illegible]

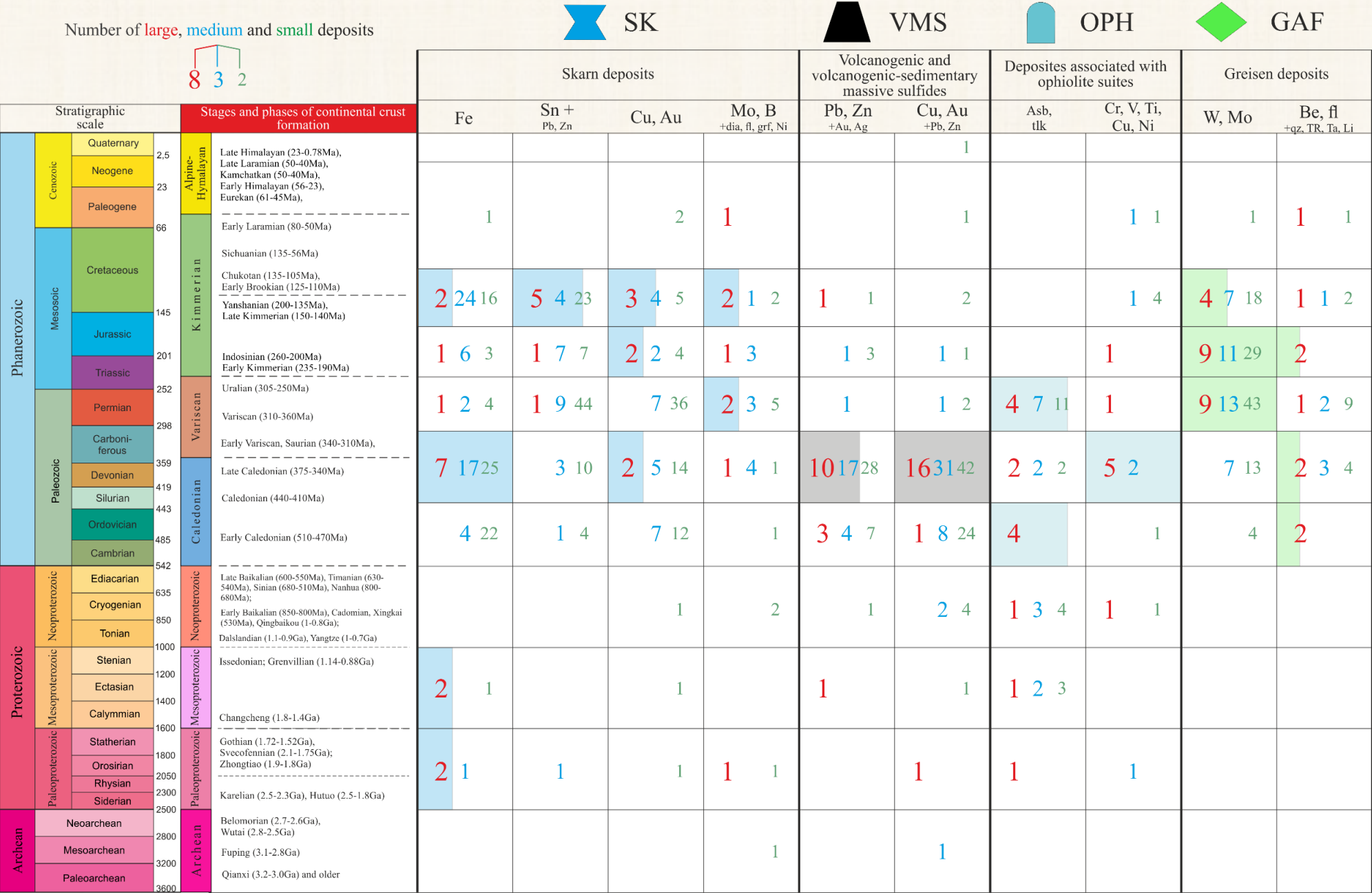
A tree diagram with a single root node at the top. Three lines branch out from the root: a red line to the left, a blue line in the middle, and a green line to the right. Below the red line is the number 8, below the blue line is the number 3, and below the green line is the number 2.



FAC

[illegible]

Metallogenic evolution of Northern, Central, and Eastern Asia



Methodology of analysis of the ore deposits formation

Evaluating the statistics of the spatial and temporal distribution of ore deposits of Metallogenic map the “**Geodynamic matching principle**” being used.

Which states:

*“If local geodynamic settings of mineral object (certain genetic type, formed within certain tectonic position) **matches** to contemporary **global geodynamic settings**, the probability of large objects formation is increased”*

Focusing on large deposits!

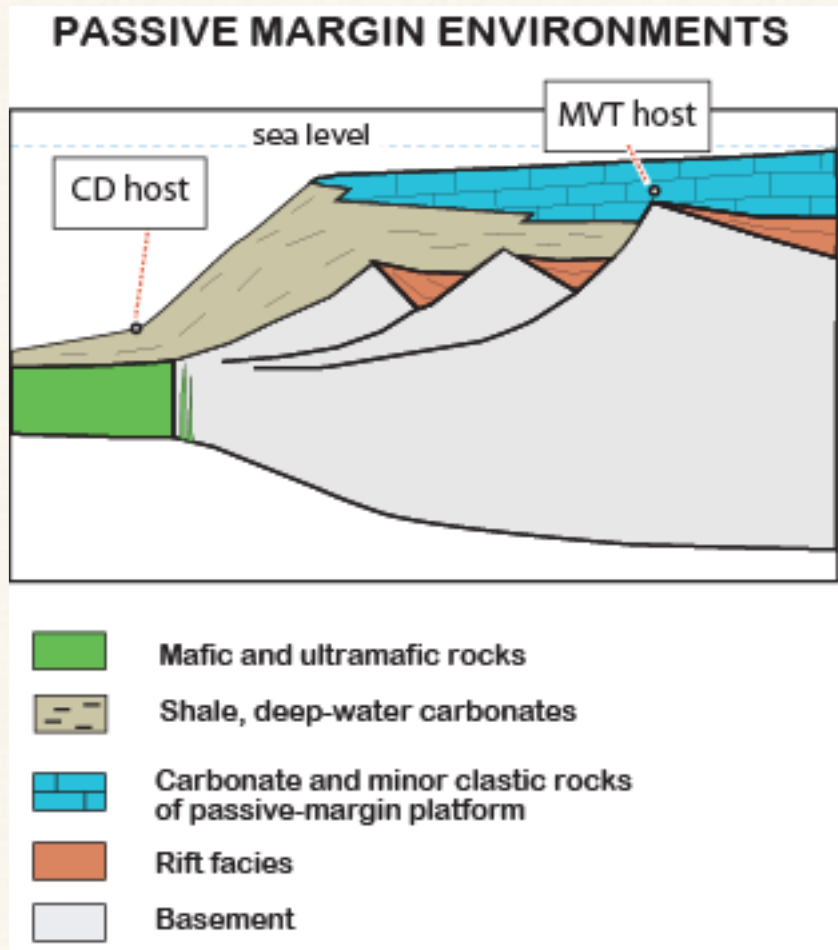
World class deposits – only 10% of all deposits in terms of contained metal, account for over 86% of Au, 79% of Ag, 84% of Cu, 71% of Zn, and 73% of Pb [D.A. Singer, 1995]

Large deposits are characterized by many parameters (lithological, tectonical, mineralogical, etc.) that are decisive in the formation of large-scale mineralization

Methodology of analysis of the ore deposits formation

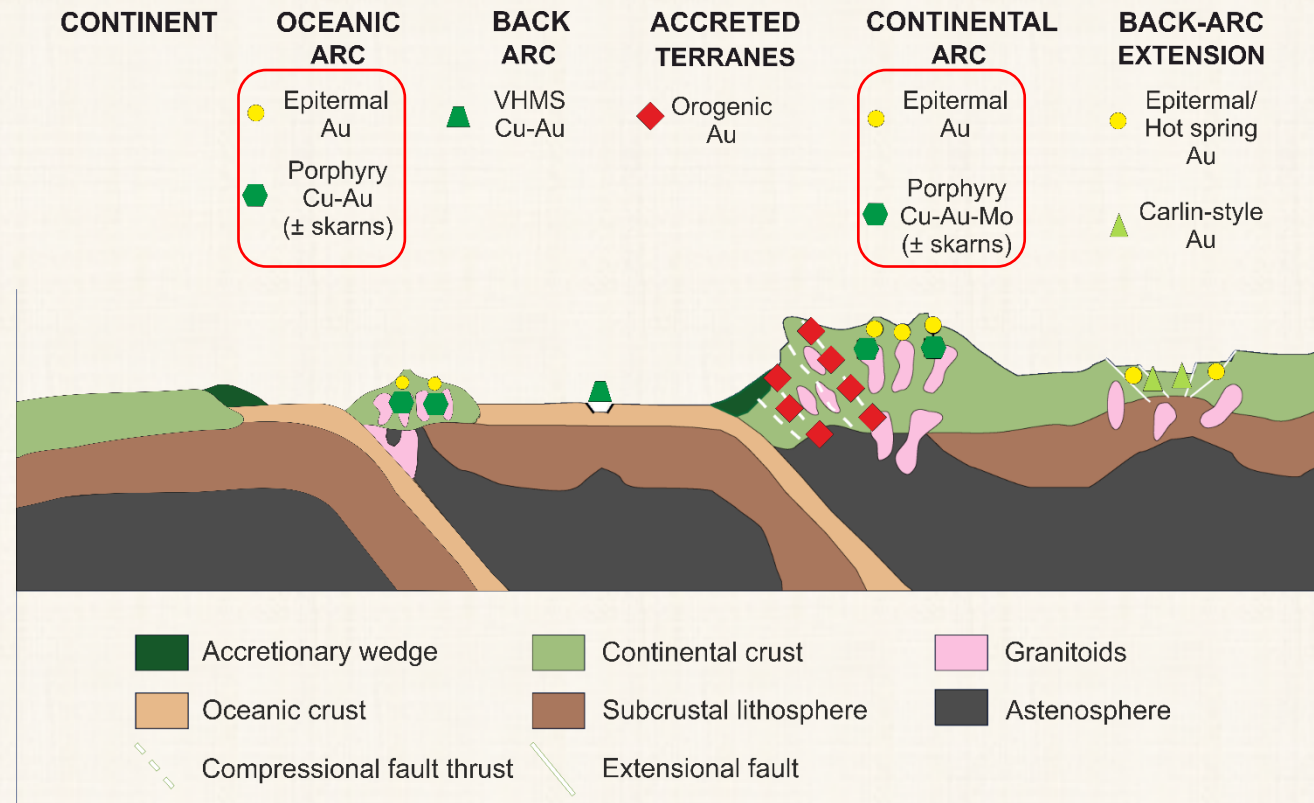
SEDEX and MVT deposits

Intraplate Tectonics and Magmatism



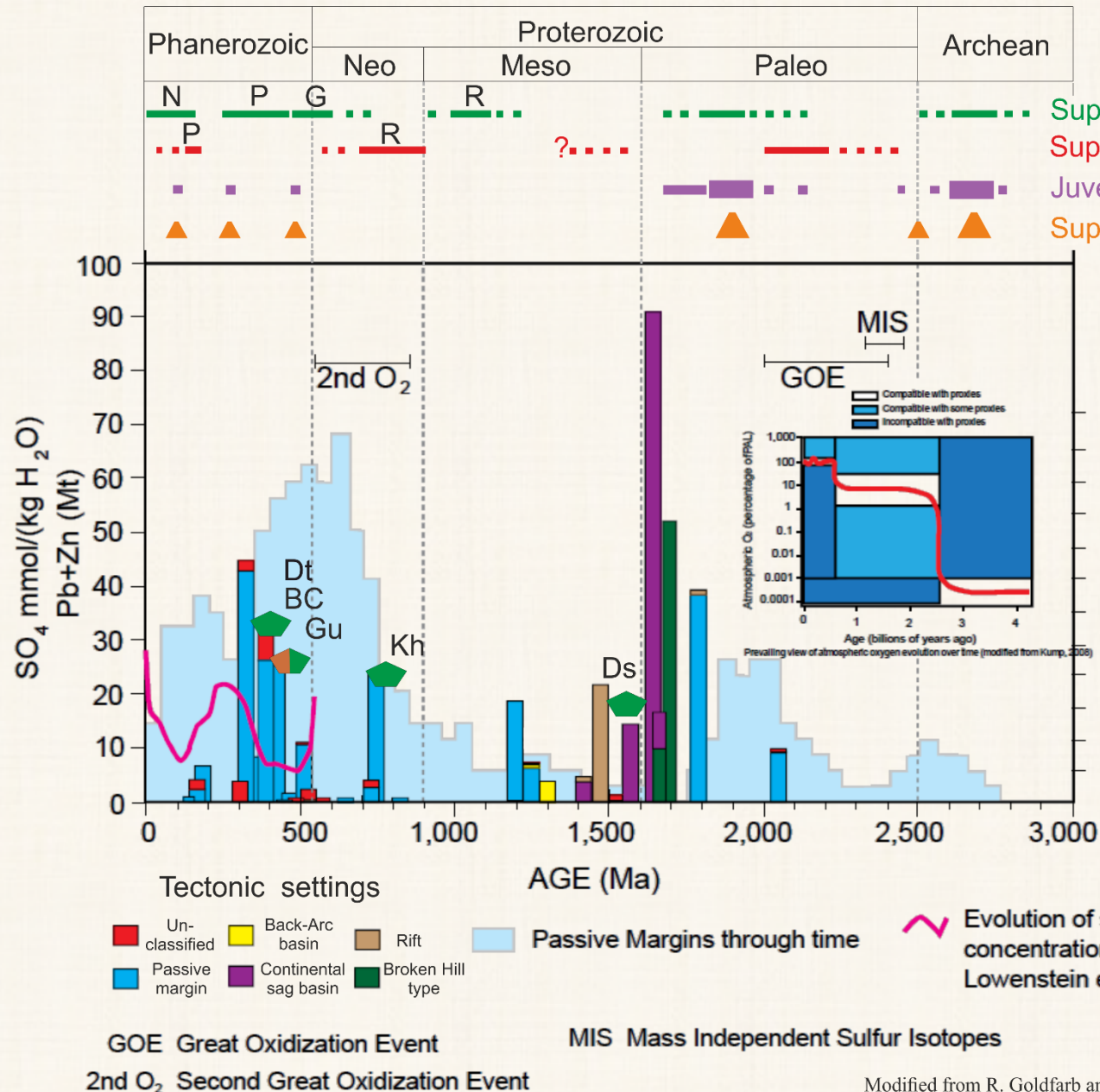
Porphyry and epithermal deposits

Accretion and Collision Structures



Secular and spatial distribution of different genetic type deposits

*Abundance of deposits =
good geodynamic conditions for
exact genetic type*



Sedimentary exhalative (SEDEX) Deposits

(also clastic-dominated sediment-hosted Pb-Zn deposits)

Ds - Dongshengmiao - Pb, Zn, Cu
Kh - Kholodninskoe - Pb, Zn
Gu - Gunduskoe - ba, Cu, Pb, Zn
BC - Bijiashan, Changba - Pb, Zn
Dt - Dachan-Tongken - Sn, Pb, Zn

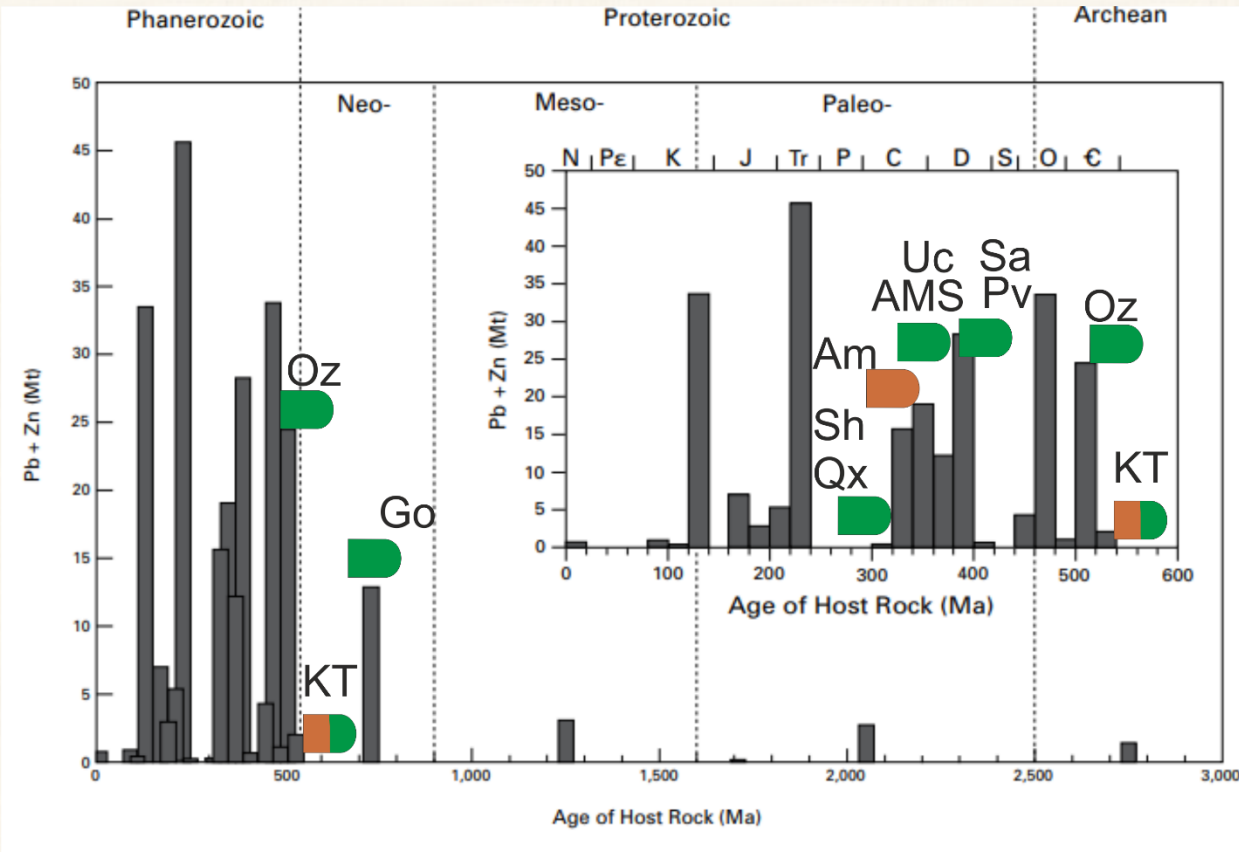
SEDEX
Pb, Zn ± Cu, Ag, Sn, ba mineralization
Barite mineralization

Secular and spatial distribution of different genetic type deposits

Abundance of deposits =
good geodynamic conditions for
exact genetic type

Mississippi-Valley type deposits (MVT)

(also known as clastic-dominated carbonate-hosted Pb-Zn deposits)



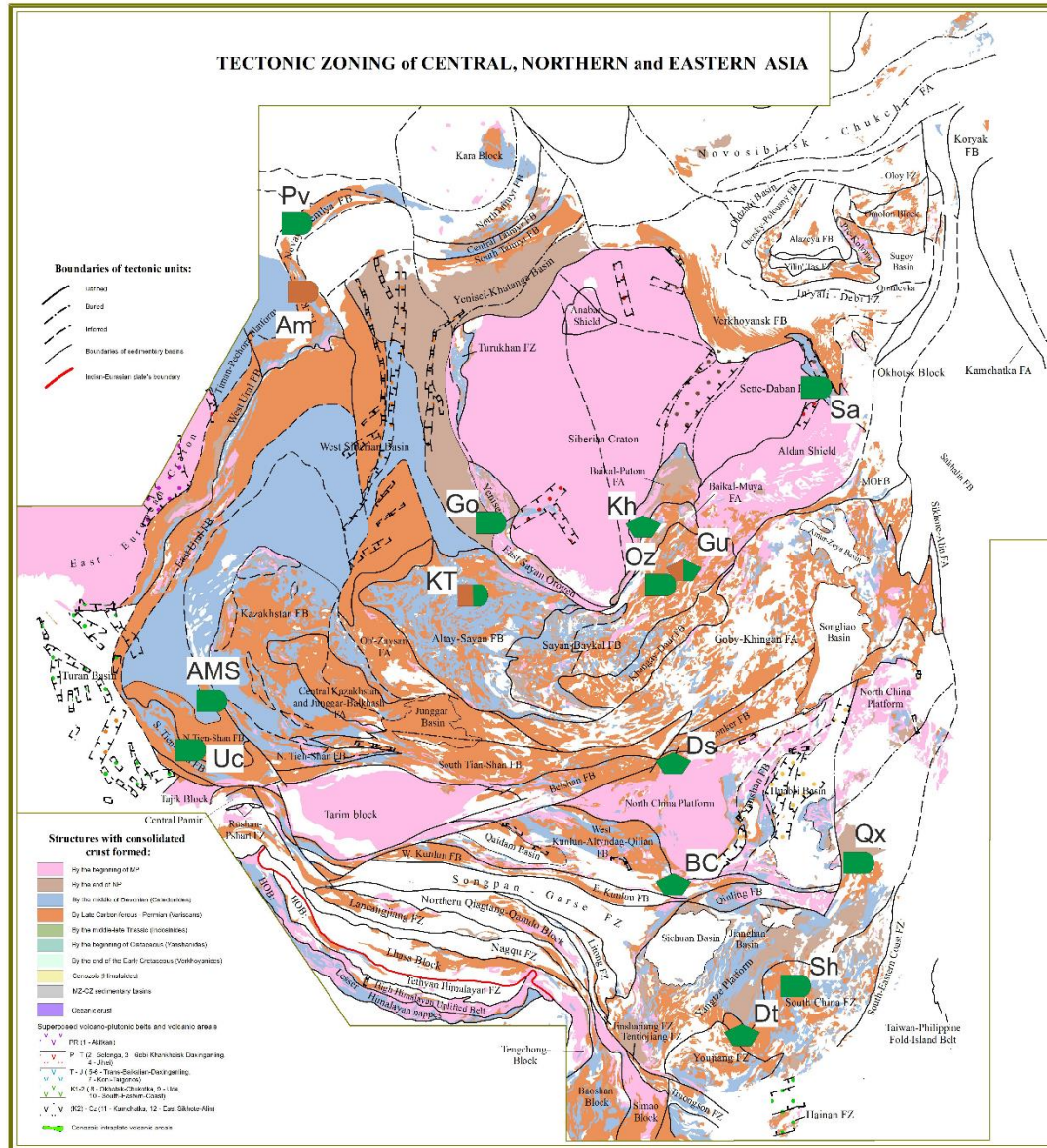
- Go - Gorevskoe - Pb, Zn, ba, Ag
- KT - Kushen'-Bulukskoe, Tolcheinskoe - ba, Pb, Zn
- Oz - Ozernoe - Pb, Zn
- Pv - Pavlovskoe - Pb, Zn, Ag
- Sa - Sardana - Pb, Zn
- Uc - Uchkulach - Pb, Zn
- AMS - Achisai, Mirgalimsai, Shalkiya - Pb, Zn
- Am - Amderminskoe - fl
- Sh - Shuikoushan - Pb, Zn
- Qx - Qixiashan - Pb, Zn

Modified from D. Leach et. al., 2010

- Pb, Zn ± Cu, Ag, Sn, ba mineralization
- Barite, fluorite mineralization
- MVT

Secular and spatial distribution of different genetic type deposits

Abundance of deposits =
good geodynamic conditions for
exact genetic type



SEDEX

Ds - Dongshengmiao - Pb, Zn, Cu
Kh - Kholodninskoe - Pb, Zn
Gu - Gunduiskoe - ba, Cu, Pb, Zn
BC - Bijashan, Changba - Pb, Zn
Dt - Dachan-Tongken - Sn, Pb, Zn

MVT

Go - Gorevskoe - Pb, Zn, ba, Ag
KT - Kushen'-Bulukskoe, Tolcheinskoe - ba, Pb, Zn
Oz - Ozernoe - Pb, Zn
Pv - Pavlovskoe - Pb, Zn, Ag
Sa - Sardana - Pb, Zn
Uc - Uchkulach - Pb, Zn
AMS - Achisai, Mirgalimsai, Shalkiya - Pb, Zn
Am - Amderminskoe - fl
Sh - Shuikoushan - Pb, Zn
Qx - Qixiashan - Pb, Zn

D MVT

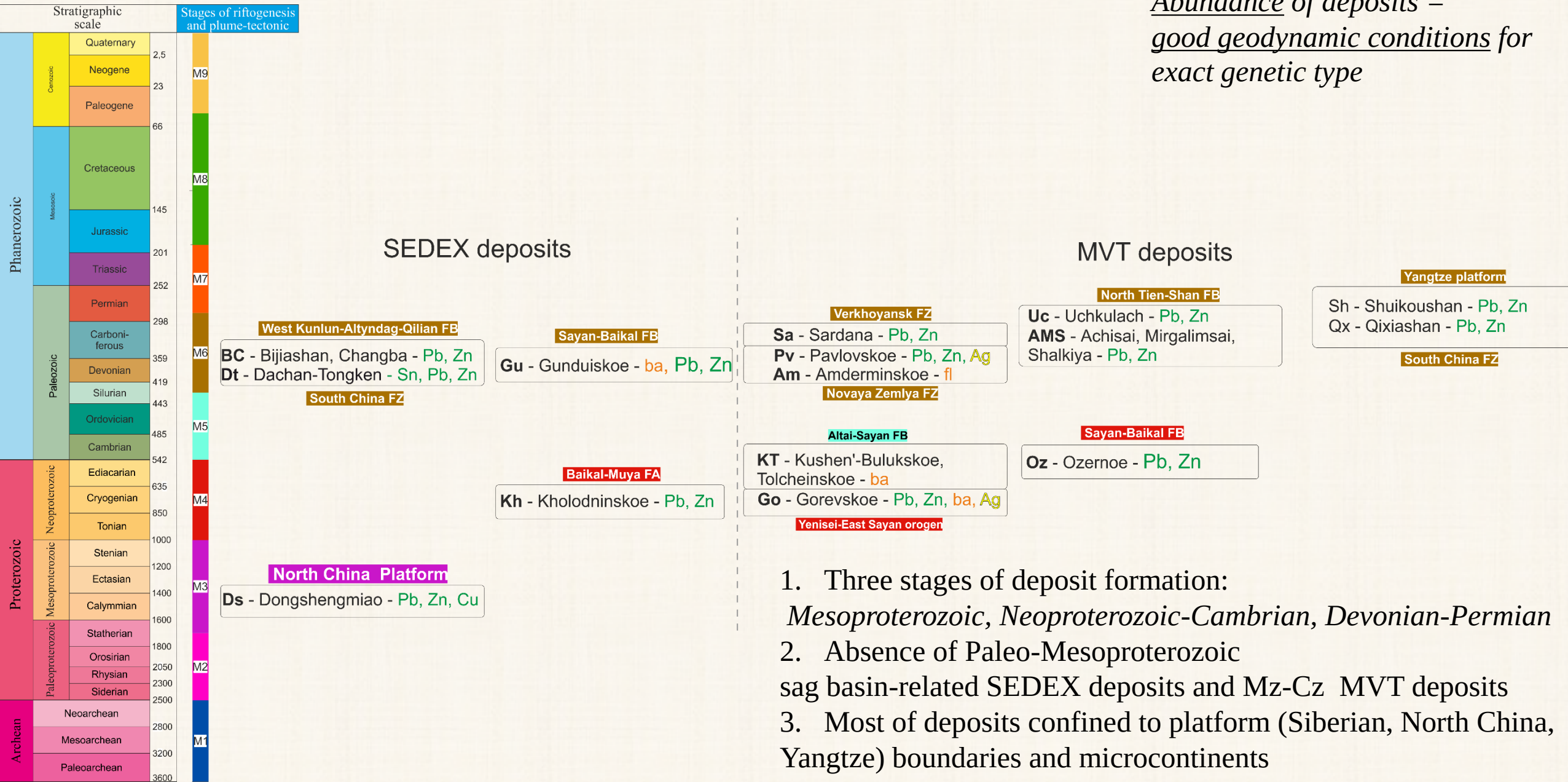


Pb, Zn ± Cu, Ag, Sn, ba
mineralization

Barite mineralization

Secular and spatial distribution of different genetic type deposits

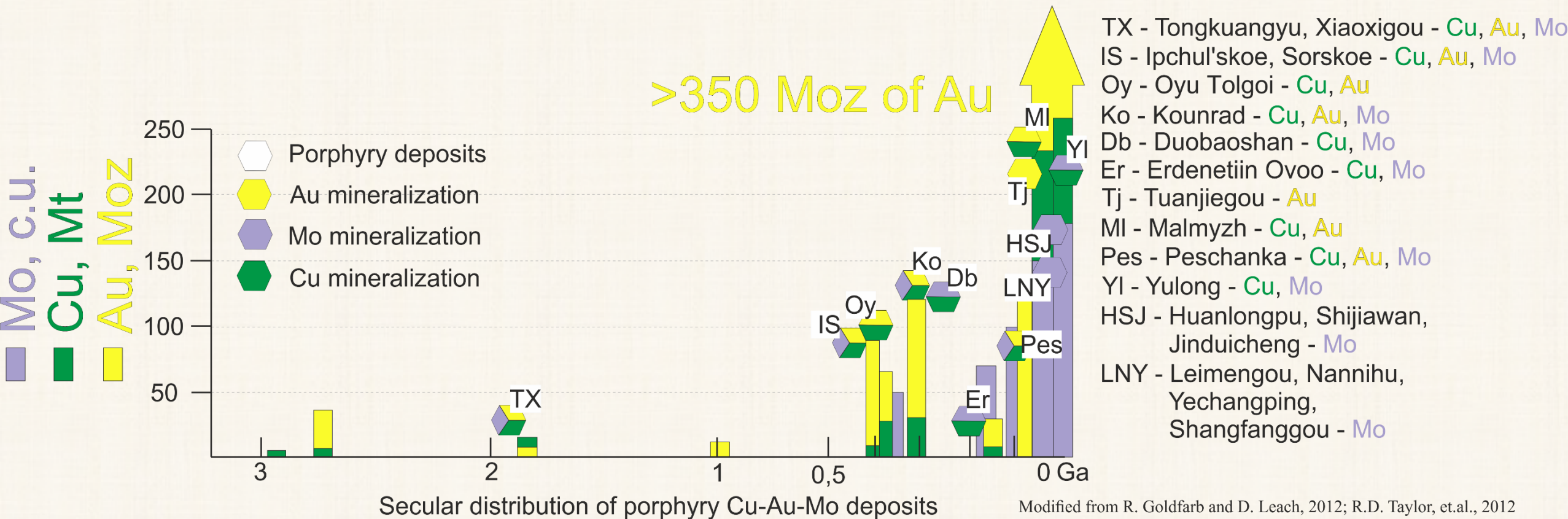
Abundance of deposits =
good geodynamic conditions for
exact genetic type



Secular and spatial distribution of different genetic type deposits

*Abundance of deposits =
good geodynamic conditions for
exact genetic type*

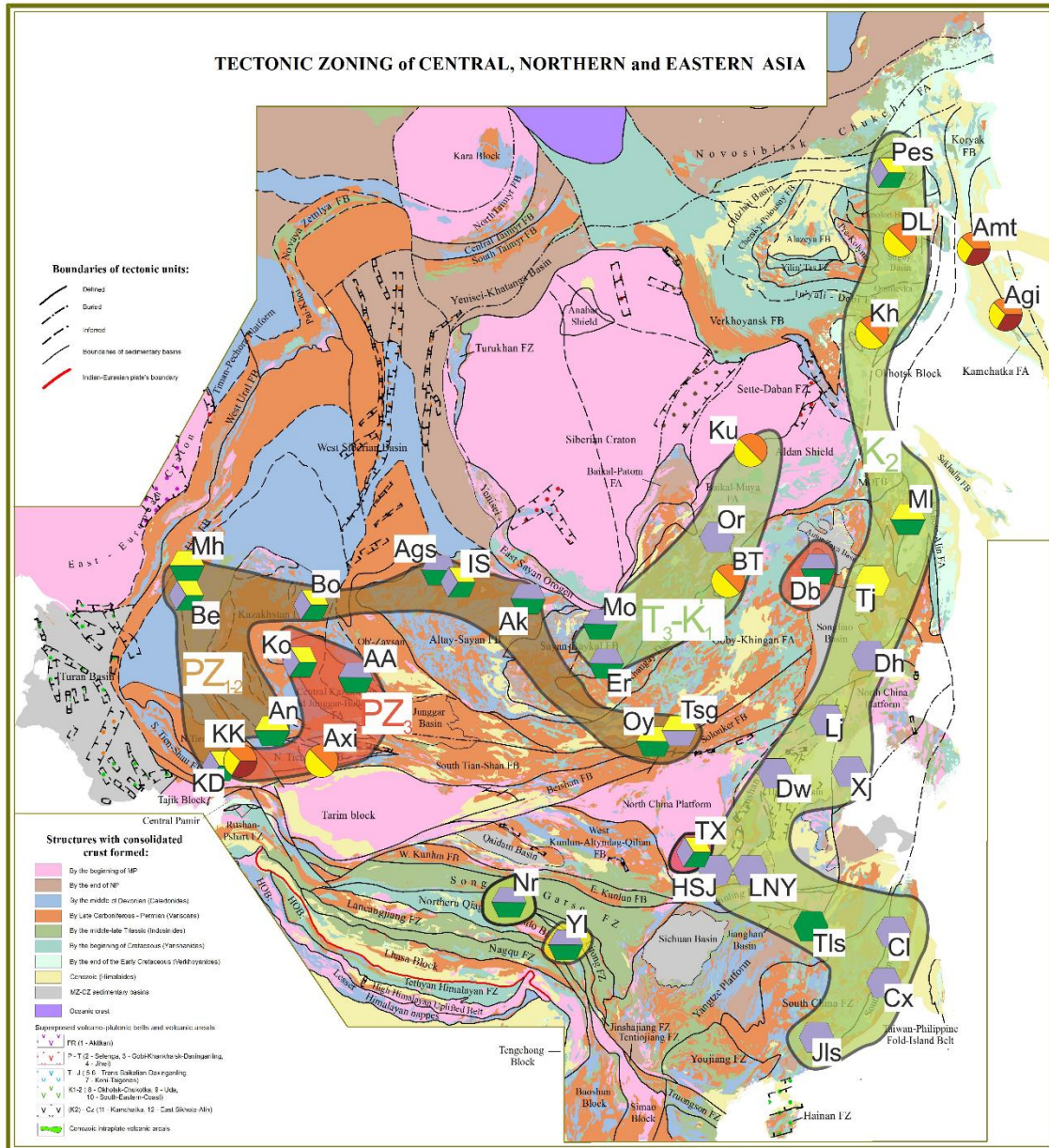
Porphyry Au-Cu-Mo deposits



Secular and spatial distribution of different genetic type deposits

Abundance of deposits =
good geodynamic conditions for
exact genetic type

TECTONIC ZONING of CENTRAL, NORTHERN and EASTERN ASIA



Epithermal

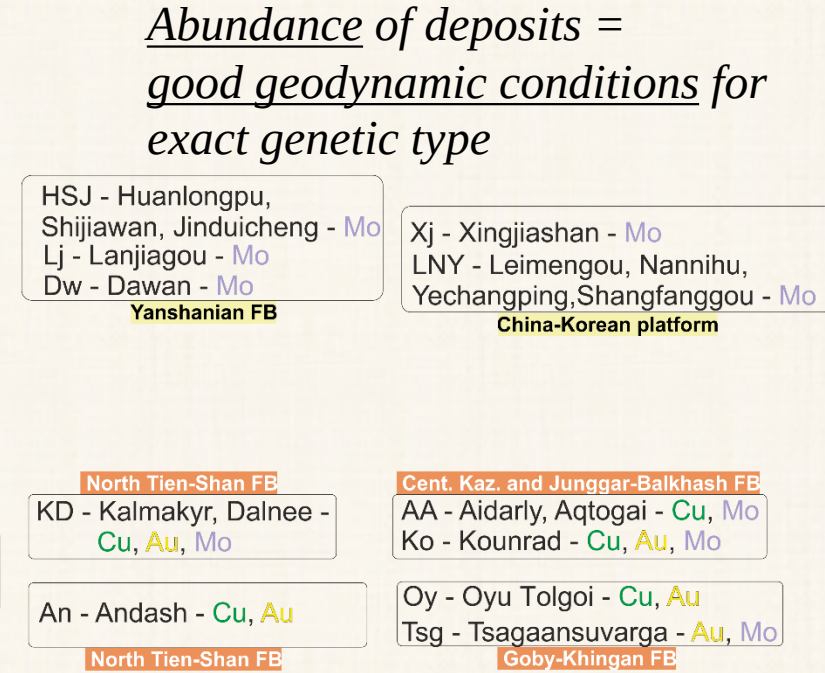
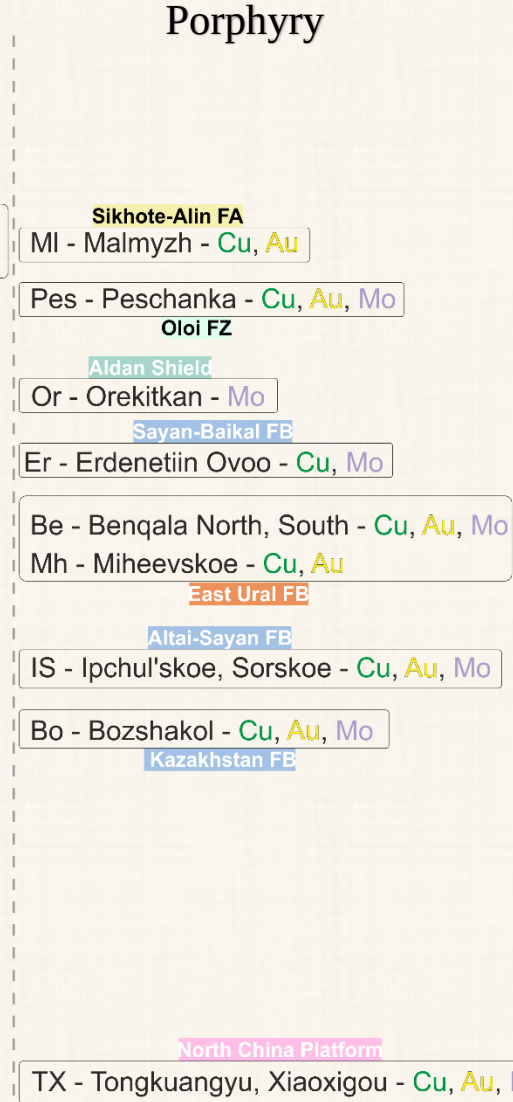
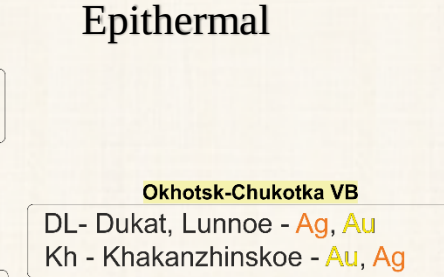
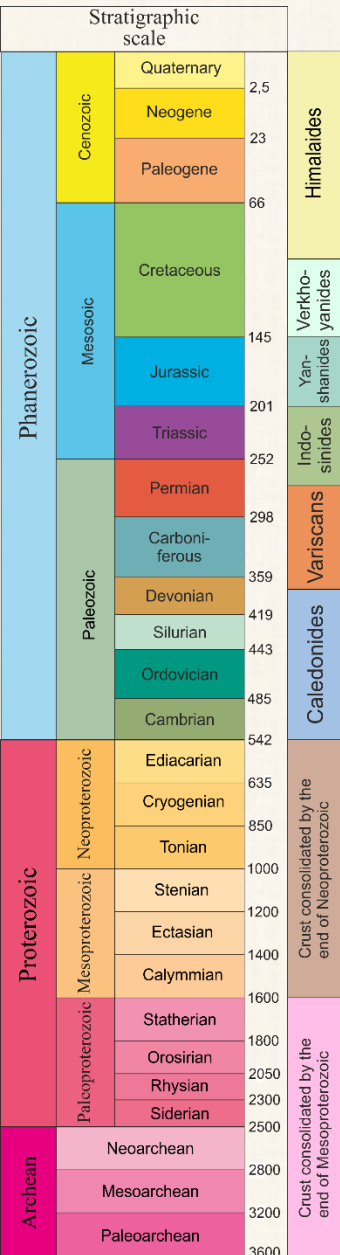
KK - Kayragach, Kochbulak - **Au, Ag, Te**
Axi - Axi - **Au, Ag**
DL - Dukat, Lunnoe - **Ag, Au**
BT - Baley, Taseevskoe - **Au, Ag**
Kh - Khakanzhinskoe - **Au, Ag**
Ku - Kuranakh - **Au, Ag**
Amt - Ametistovoe - **Au, Ag, Te**
Agi - Aginskoe - **Au, Ag, Te**

Porphyry

TX - Tongkuangyu, Xiaoxigou - **Cu, Au, Mo**
Bo - Bozshakol - **Cu, Au, Mo**
IS - Ipchul'skoe, Sorskoe - **Cu, Au, Mo**
Oy - Oyu Tolgoi - **Cu, Au**
Be - Benqala North, South - **Cu, Au, Mo**
An - Andash - **Cu, Au**
Tsg - Tsagaansuvarga - **Au, Mo**
Mh - Miheevskoe - **Cu, Au**
Ags - Agaskyrskoe - **Cu, Au, Mo**
KD - Kalmakyr, Dalnee - **Cu, Au, Mo**
Ak - Aksukskoe - **Cu, Mo**
Ko - Kounrad - **Cu, Au, Mo**
Er - Erdenetiin Ovoo - **Cu, Mo**
Mo - Malo-Oinogorskoe - **Cu, Mo**
Or - Orekitkan - **Mo**
Pes - Peschanka - **Cu, Au, Mo**
MI - Malmyzh - **Cu, Au**

Db - Duobaoshan - **Cu, Mo**
AA - Aidarly, Aqtogai - **Cu, Mo**
Tj - Tuanjiegou - **Au**
Dh - Daheishan - **Mo**
Lj - Lanjiagou - **Mo**
Xj - Xingjiashan - **Mo**
Dw - Dawan - **Mo**
HSJ - Huanlongpu, Shijiawan, Jinduicheng - **Mo**
LNY - Leimengou, Nannihu, Yechangping, Shangfanggou - **Mo**
Tls - Tonglushan - **Cu**
Cl - Chilul - **Mo**
Cx - Caixi - **Mo**
Jls - Jilongshan - **Mo**
Nr - Narigongma - **Cu, Mo**
Yl - Yulong - **Cu, Mo**

Secular and spatial distribution of different genetic type deposits



Abundance of deposits =
good geodynamic conditions for
exact genetic type

1. Three main stages of epithermal deposits formation are spatially and temporarily separated: *Carbon-Permian, Early Cretaceous, Late Cenozoic*
2. *Paleozoic* deposits are mostly within western and central parts of CAO B
3. Porphyry and epithermal deposits spatially and temporarily correlate

Conclusion

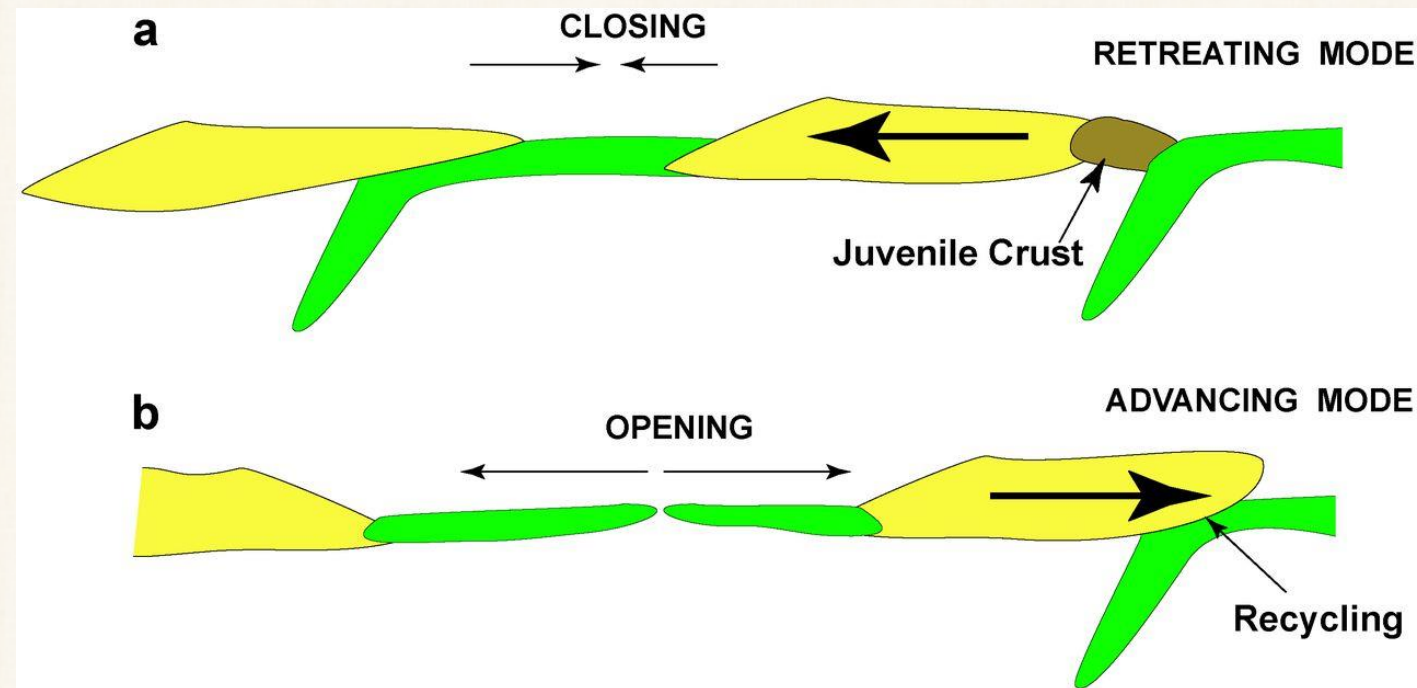
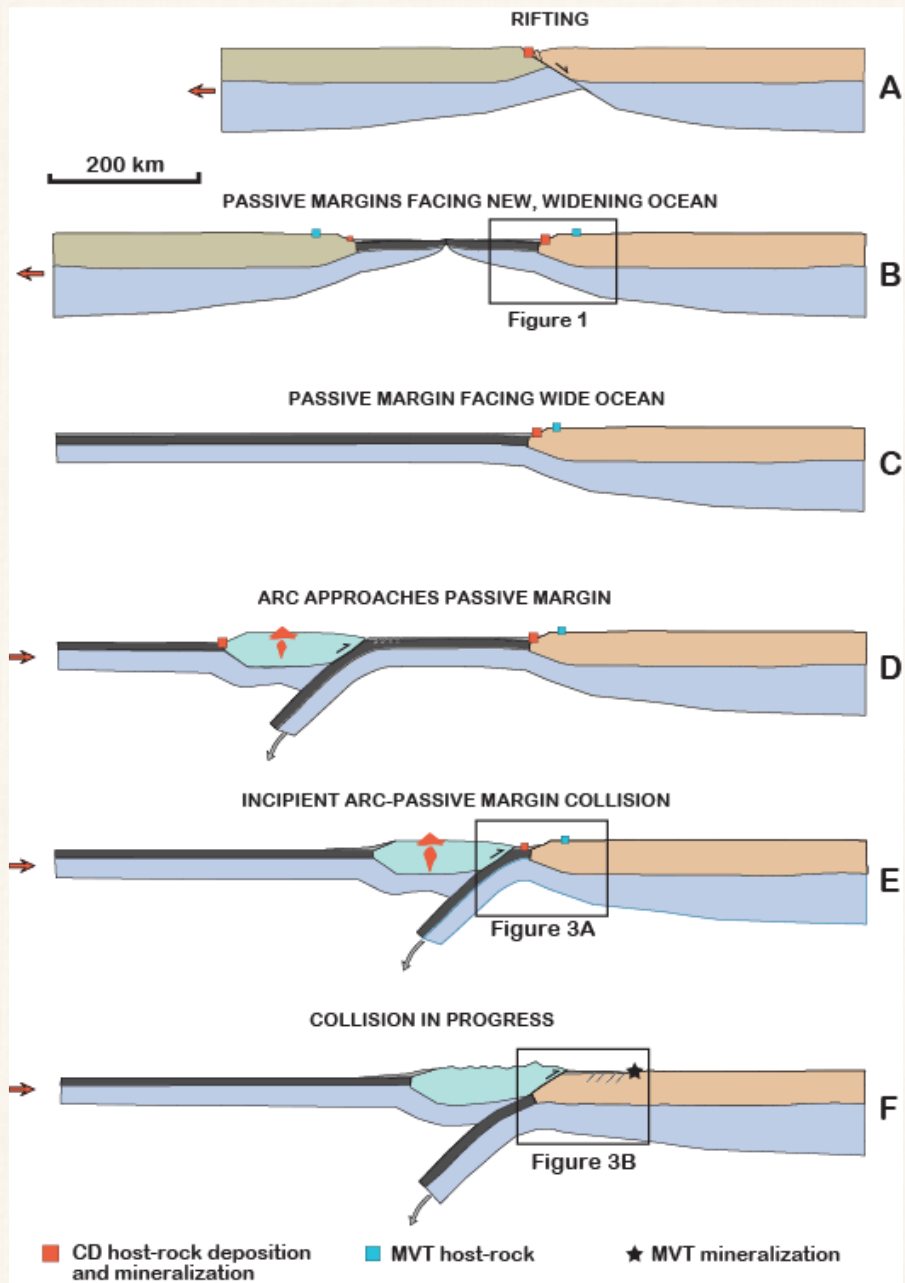
- General work for Explanatory Note in its final stage. Major revision is almost finished.
- Detailed metallogenic analysis for the Map and its database is being provided
- In the framework of the methodology, 4 genetic types of deposits were studied that are promising (SEDEX, MVT) and most productive (Epithermal, Porphyry) deposits for the project area. This work could improve the prospects of new large deposits.

Decisive results of the analysis and synthesis of metallogenic data are:

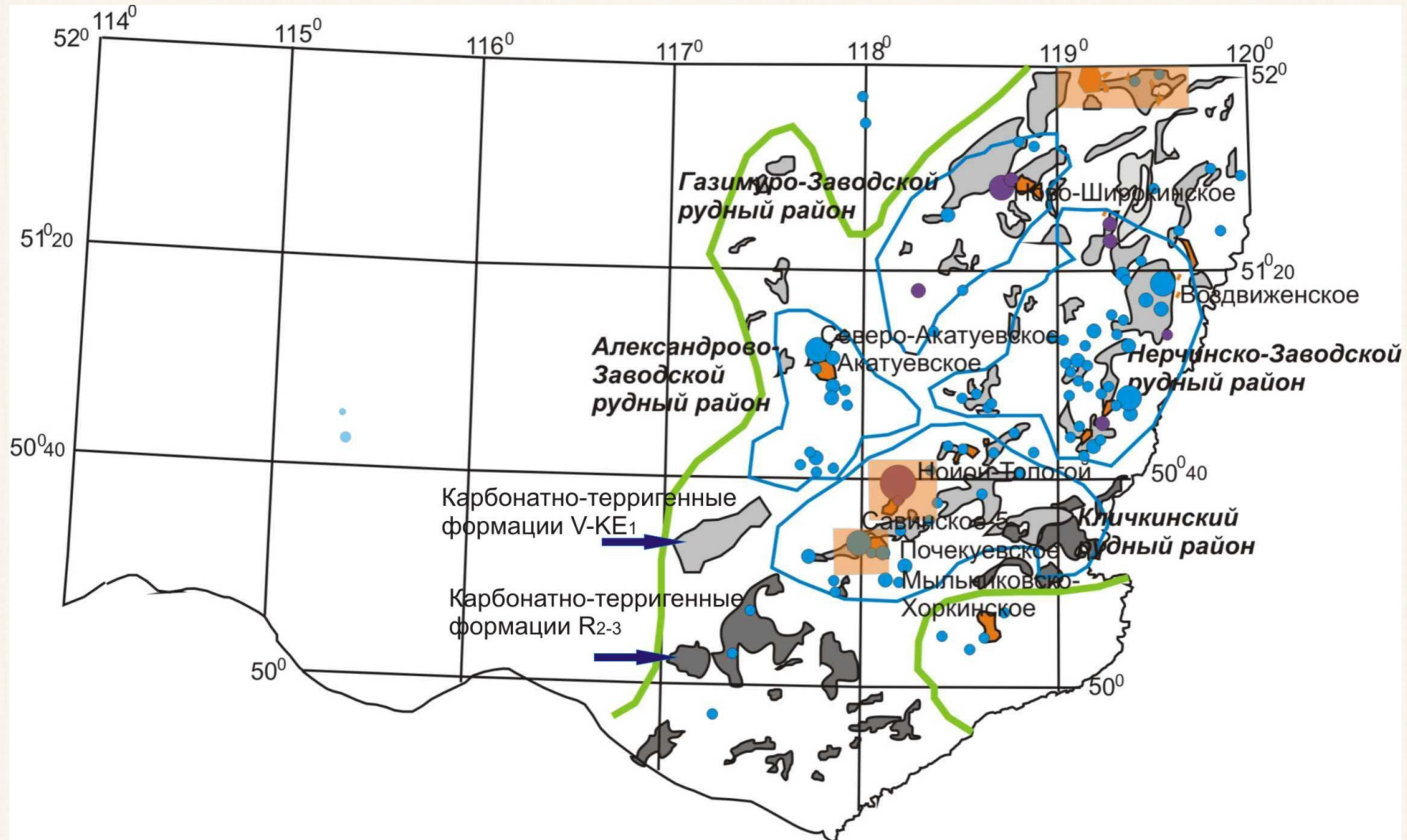
- to continue the studies in the same methodology of remaining genetic types of mineral deposits with including of the results to Explanatory Note for the Metallogenic map
- to complete the Explanatory Note compilation, editing and publication
- to submit published Metallogenic Map and Explanatory Note on the 36th IGC (India, 2020)

Thank you for your attention!





Boundary of the Argun polymetallic belt within sheets M-49,50 is drawn in accordance with the distribution area of specialized for Pb, Zn carbonate-terrigenous formations V-KE1, formed in a global epoch of shelf facies (transgression) in the **northern boundary of Gondwana**.



Eastern Transbaikalia, sheets M-49,50.

Areas of tectonophysical extension in the epoch of dispersed rifting K₁ (transgression start) are traced by sedimentary-volcanogenic depressions

